Smart Procurement AI Assistant – Supplier RFX & Risk Analysis

Sustainable Packaging Components

Jahnavi Srinag | AI/ML B | 06/11/2025
PES1UG23AM122

Overview: AI in Procurement

- • AI helps automate RFX creation and supplier evaluation.
- • Reduces manual workload and increases decision accuracy.
- • Enables data-driven supplier risk scoring.

RFP Summary – Sustainable Packaging Components

- • Supplier Criteria: ISO 14001 compliance, recyclability focus.
- • Evaluation: Cost, Quality, Delivery, ESG Score, Financial Stability.
- • Key ESG Questions:
- 1. % recyclable materials?
- 2. Carbon footprint per ton?
- 3. Sustainability reporting standards?
- 4. Waste & water control measures?
- 5. Environmental certifications?

Dataset Overview

- • Cost_Index – Lower = better.
- • Quality_Score – Higher = better.
- • Delivery_Reliability – On-time performance.
- • ESG_Score – Sustainability level.
- • Financial_Stability – Long-term health.
- • Overall_Risk – Target label.

Weka Decision Tree Output

Test options

- Use training set
- Supplied test set — Set...
- Cross-validation — Folds 10
- Percentage split — % 66

More options...

(Nom) Overall_Risk

Start | Stop

Result list (right-click for options)

- 09:33:53 - trees.J48
- 09:38:23 - trees.J48
- 09:41:42 - trees.REPTree
- 09:43:26 - trees.REPTree

Classifier output

```
Time taken to build model: 0.35 seconds

=== Stratified cross-validation ===
=== Summary ===

Correctly Classified Instances        8471               84.71   %
Incorrectly Classified Instances      1529               15.29   %
Kappa statistic                          0.7582
Mean absolute error                      0.1373
Root mean squared error                  0.2874
Relative absolute error                 32.525  %
Root relative squared error             62.5461 %
Total Number of Instances           10000

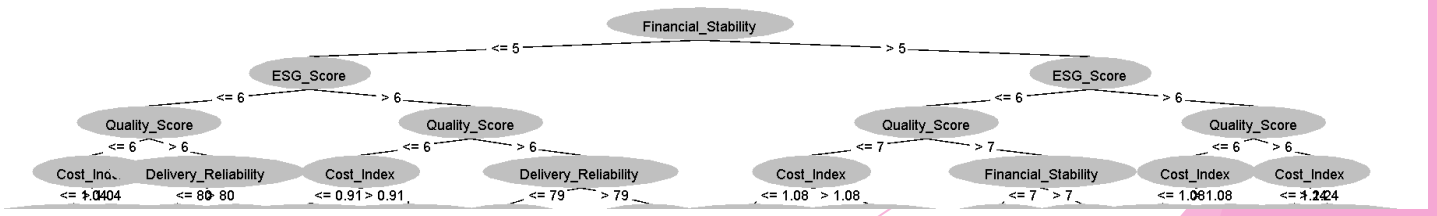
=== Detailed Accuracy By Class ===

                 TP Rate   FP Rate   Precision   Recall  F-Measure  MCC      ROC Area  PRC Area  Class
                 0.816     0.132     0.805      0.816     0.811      0.683     0.896     0.811     Mediu
                 0.892     0.075     0.894      0.892     0.893      0.818     0.960     0.937     Low
                 0.813     0.037     0.833      0.813     0.823      0.784     0.960     0.867     High
Weighted Avg.    0.847     0.091     0.847      0.847     0.847      0.757     0.934     0.873

=== Confusion Matrix ===

    a    b    c   <-- classified as
 3278  437  300 |    a = Medium
  447 3693    0 |    b = Low
  345    0 1500 |    c = High
```

Tree View



Insights from Analysis

- • Low Delivery Reliability + Low ESG Score → High Risk.
- • High Quality Score + Financial Stability → Low Risk.
- • ESG and reliability influence risk more than cost.

Recommendation

- • Choose suppliers with strong ESG, reliable delivery, and stable finances.
- • Accept slightly higher cost for long-term reliability and sustainability.

REQUEST FOR PROPOSAL (RFP)

pes1ug23am122

Eco-Friendly Electronic Sensor Components

1. Company Overview & Requirement

EcoTech Manufacturing is seeking suppliers of eco-friendly electronic sensors (temperature, humidity, motion, etc.) that comply with environmental regulations and support sustainable production. Suppliers must ensure low-toxicity materials and energy-efficient design.

2. Supplier Eligibility Criteria

- Minimum 3 years experience in sensor manufacturing.
- Compliance with RoHS and REACH.
- Environmental management system (ISO 14001 preferred).
- Quality system certification (ISO 9001 or equivalent).
- Ability to provide sustainability and material compliance documentation.

3. Evaluation Criteria (Weighted Score: 100)

4. ESG Performance Questions

- What % of materials used in your sensors are recyclable or low-impact?
- Provide carbon emissions data per unit/batch produced.
- Which sustainability standards or reports do you follow (GRI, CDP, etc.)?
- Describe waste, water, and hazardous material reduction efforts.
- List environmental & labor safety certifications (ISO 14001, SA8000).

5. Required Submissions

- Company profile & capabilities
- Product datasheets & compliance certifications
- Material & sustainability documentation
- ISO certificates (9001/14001)
- Financial statements (last 2 years)
- Answers to ESG questions

6. Submission Deadline

Submit proposals to procure

EcoTech Manufacturing reserves the right to shortlist, negotiate, or decline proposals.

RF

Criteria	Description	Weight
Technical Capability	Performance & specification match	25
Quality Assurance	Testing & failure rate control	20
Delivery & Reliability	On-time logistics & lead time	15
Cost Competitiveness	Pricing & cost stability	15
ESG & Sustainability	Environmental & ethical practices	15
Financial Stability	Business continuity & credit health	10

Conclusion

- • AI speeds RFP drafting and supplier risk assessment.
- • Decision Tree reveals the strongest risk factors.
- • Data-driven supplier selection improves outcomes.